

Risk Assessment and Mitigation

ENG1 Cohort 1 Team 3

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A risk is anything that can threaten the project, often referring to future events that may or may not end up happening, that can be both known and unknown.

In this process, we started by identifying all the possible risks that could affect the development of the game. We outlined several sections of the project, such as the requirements, and as a team, we brainstormed potential risks for each section, noting them down accordingly. We then eliminated risks that we deemed extremely unlikely or inconsequential for a project of this scope. For the remaining risks, we created simple yet clear descriptions to ensure that when we revisited them later, we could easily understand what each risk entailed. Splitting the risks into different types/sections helps us easily identify what each risk corresponds to and reference them when we are working on that part of the project.

After identifying all our initial risks we analysed each risk as a team and discussed their **severity** and **likelihood**. We decided that the likelihood would span from Low to Medium to High as we figured that this would be enough detail for the likelihood of each risk. Additionally, we decided to categorise the severity of each risk as Low, Medium and High for the same reasons. Having analysed each risk, we then developed a plan to address each one. This plan had to include at least one way to avoid or mitigate each risk and if possible both. We thought that having both an avoidance and mitigation strategy for a risk was useful as we had a way to reduce the chance of the risk occurring and the impact it would have if it did occur. For certain risks, we also developed a contingency plan which is the best course of action if the risk isn't avoided or mitigated and often entails falling back to earlier proven solutions.

Lastly, we assigned an owner to each risk. The owner's responsibility was to assess the likelihood and severity of the risk at least once a week. Additionally, owners were required to report any changes to the team during our weekly meetings. This allowed the team to collectively decide whether the risk's values needed to be updated and if any strategies to address the risk needed to be revised or reinforced. The assignment of risks was based on each team member's primary role and the type of risk most closely associated with that role. For instance, the project lead was assigned risks categorised as 'project' risks. This ensured that the owner of the risk was familiar with what the risk meant and its effects, meaning that they will be better able to monitor that risk over the course of the project.

All this above information about each risk was finally input into our risk register which is a table containing every risk along with their:

ID, Type, Description, Likelihood, Severity, Avoidance/Mitigation/Contingency Plan, and Owner. This register allows our team to easily lookup any risk and have all the information about that risk clearly displayed to them. The Likelihood and Severity columns were color-coded, ranging from green for lower values to red for higher values, ensuring the information was even clearer for our team.

When there is an update to a likelihood or severity rating, it will be documented in the following format:

Likelihood Severity <Date> <Reason for Change>

Risk Register

ID	Type	Description	Likelihood	Severity	Avoidance/Mitigation/Contingency Plan(s)	Owner
R1	Project	One or more members of our team are unable to participate	Medium	High	Mitigation: Make sure our bus factor remains at least 2 by having at least two people with complete knowledge of any part of the code, documentation, and the write up.	Mansi
R2	Software	Use of poor quality Libraries	Low	Medium	Avoidance: Perform careful research of third party libraries before implementing them into our code. Ensure libraries are well tested and reasonably popular to minimise risk.	Harry
R3	Product	Code structure and readability reduce as the project progresses.	Low	High	Avoidance: Ensure project architecture is well planned and code is reviewed before the final version is committed.	James
					Mitigation: Comply with relevant style guides and ensure code is well documented.	
R4	Software & Hardware	Product is unable to be built/run on the required hardware or Operating Systems	Medium	High	Avoidance: Check with our customer if they have specific requirements for device specification or operating systems the project must be compatible with.	Vadim
					Avoidance: Periodically review the performance of the project using benchmarking and ensure CI systems attempt to build the project for multiple OS targets.	
R5	Requirements	Requirements that are written don't correspond to what the customer wants	Medium	High	Avoidance: Meet regularly with the customer and specifically discuss the user level requirements with them. Then carefully create the system requirements as a team based on the user requirements that have been gathered.	Laina
R6	Requirements	The customer changes the requirements causing us to have to rewrite large parts of	Medium	High	Mitigation: Ensure that the architecture of the project is modular and when designing it, ensure the possibility of expansion and adding features is maintained.	Almira

ID	Type	Description	Likelihood	Severity	Avoidance/Mitigation/Contingency Plan(s)	Owner
		the game, possibly under limited time			Mitigation: Use an agile software development methodology, which will allow for flexible response requirements changes late into the project. This encompasses meeting with the customer frequently, which should reduce the number of the changes that arise later on.	
					Contingency: If there is not enough time to make the changes or the changes required are closer to a prior version we should fallback to that prior version of the solution.	
R7	Project	Misalignment between member's strength and role	Low	Medium	Avoidance: Early on in the project we should try to determine our team members strengths and assign roles to them that best fit those strengths.	Vadim
					Mitigation: Assigning more than one member to the same role to leverage different skills.	
R8	Requirements	Unclear or ambiguous requirements	Low	High	Avoidance: Make sure requirements are carefully written from the start.	Almira
					Mitigation: If a team member feels they are ambiguous or unclear, make sure to clarify this with the rest of the team and the customer if necessary.	
R9	Software & Hardware	Use of tools that are overcomplicated or only one team member understands	Low	Medium	Avoidance: Have discussions between the team about which tools we've used before when choosing one.	Harry
					Mitigation: If a new tool is chosen and it is complex or some team members are unfamiliar with it, hold a meeting to show all team members the basics.	
R10	Project	Poor time management and missing the deadline	Low	High	Avoidance: Prioritise working on the fundamental requirements of the project before spending time on extra features.	Mansi

ID	Type	Description	Likelihood	Severity	Avoidance/Mitigation/Contingency Plan(s)	Owner
					Mitigation: Using a gantt chart from the start, allows time buffers between tasks to accommodate sudden events.	
R11	Project	Lack of communication within team causes issues	High	Medium	Avoidance: Set up a quick and easy communication system for the team to use.	Mansi
					Mitigation: Team members should clarify if they have any questions or misunderstandings.	
R12	Project	Project schedule is not defined or understood.	Medium	High	Avoidance: Create a clear project schedule and assign tasks to each team member.	Almira
					Mitigation: Hold regular meetings with the team to check on task progress and ensure new tasks are set.	
R13	Project	Scope of project increases and results in some requirements not being met in time.	Medium	High	Avoidance: Clearly follow the project schedule and ensure that all requirements are first met before extra features are discussed or worked on.	Vadim
R14	Write Up	Parts of the writeup are poorly formatted or hard to understand.	High	Low	Avoidance: Keep the format of the writeup segments fairly consistent throughout and make sure the format is easy to understand, e.g. paragraphs should contain sentences of information that links together.	Laina
					Mitigation: Ensure that all parts of the writeup are proofread by the writer and at least one other member of the team.	
R15	Project	Loss of key files or deliverables.	Low	High	Avoidance: Use GitHub and google drive. Regularly create backups of important documents and pages.	Vadim
R16	Hardware	Issues arising when running the game on different platforms.	Medium	High	Avoidance: Conduct cross-platform testing and utilise platform-independent tools.	James

R17	Software & Hardware	Unexpected crashing of game during testing.	High	Medium	Mitigation: Maintaining error logs, conducting thorough code reviews, implementing automated tests and regularly gathering feedback.	Harry
R18	Estimation	Game instructions are too complicated or unclear for users.	Low	High	Mitigation: Provide instructions or a tutorial at the beginning of the game and display the controls on screen.	Mansi
R19	People	The game's content is sensitive to cultural issues.	Low	High	Avoidance: Thoroughly research all uncertain areas and conduct user testing with diverse groups to gather feedback and make necessary adjustments.	Almira
R20	People	The game is not adequately accessible (lack of output) for a wide audience.	Medium	High	Mitigation: Implement inclusive design principles, ensure adequate audio and visual feedback, provide options for customisation (such as subtitle adjustments and colour-blind modes), and conduct user testing with diverse groups to gather feedback and make necessary adjustments	Laina
R21	Product	The game violates intellectual property rights.	Low	High	Mitigation: Use original code, include non-copyrighted content while being aware of relevant trademarks.	James